



Facet-tunable coral-like Mo₂C catalyst for electrocatalytic hydrogen evolution reaction

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ABSTRACT

Developing efficient and stable transition metal-based electrocatalysts for the hydrogen evolution reaction (HER) in alkaline media remains a great challenge due to slow water dissociation kinetics. Here, a novel coral-like porous nitrogen-doped Mo₂C electrocatalyst (3D Mo₂C) derived from ZIF-8 doped with phosphomolybdic acid hydrate (POM) was proposed for the electrocatalytic hydrogen evolution reaction via facet regulation. Density functional theory (DFT) calculations clarified that the Mo₂C (0 0 2) facet accelerated water dissociation, while the Mo₂C (1 0 1) facet more easily adsorbed H* to generate H₂. Thus, Mo₂C facets (1 0 1) and (0 0 2) with a suitable ratio showed synergistic effects to exhibit excellent alkaline HER performance. The coral-like porous Mo₂C exposed sufficient active sites and facilitated electron and mass transfer to accelerate alkaline HER. The 3D Mo₂C (1:1) catalyst exhibited excellent HER activity with a low overpotential of 110 mV and a small Tafel slope of 73.9 mV dec⁻¹ in 1 M KOH.

1. Introduction

Hydrogen energy is a clean and renewable energy source that can alleviate the increasingly serious energy crisis and environmental problems [1,2]. Water electrolysis using wind or solar power provides an efficient and low-cost approach for hydrogen production [3]. Compared with acidic and neutral electrolytes, the alkaline electrocatalytic hydrogen evolution reaction (HER) is more desirable due to its lower corrosivity towards equipment and transition metal electrodes [4,5]. In alkaline environments, H* production requires an additional dissociation process compared with acidic electrolyte known as the Volmer step (H₂O + e⁻ → H* + OH⁻) [6]. Platinum (Pt)-based electrocatalysts exhibit excellent hydrogen evolution performance with near-zero Gibbs free energy (ΔG_{H*}) under alkaline conditions [7]. However, the scarcity and high costs of Pt limit their large-scale applications. Therefore, it is crucial to develop efficient and stable non-precious metal alkaline HER electrocatalysts.

Transition metal carbides exhibit excellent performance and good stability during the alkaline HER, especially molybdenum carbides (Mo_xC), which have a similar electronic structure to Pt-group metals [8–10]. However, the preparation of Mo_xC requires extremely high temperatures, which inevitably leads to the aggregation of nanoparticles

and a reduction in the number of exposed active sites, resulting in low HER activity [11]. Excessively strong Mo-H bonds make it difficult to desorb H* from the Mo_xC surface during the alkaline HER [12]. Thus, it is necessary to determine how to improve the hydrogen evolution properties and control the agglomeration of Mo_xC electrocatalysts.

The electrocatalytic performance of Mo_xC for HER is mainly related to the number of exposed catalytic sites and effective crystal planes. In recent years, metal–organic frameworks (MOFs) have stood out in the field of catalysis due to their specific structural properties, large specific surface areas, and controllable structures [13–15]. The abundant organic ligands and porous structure of MOFs can prevent the agglomeration of Mo_xC nanoparticles during high-temperature calcination [16]. For example, Lou et al. [9] synthesized Mo_xC octahedral nanoparticles by encapsulating polyoxometalates (POMs) in a Cu metal–organic framework (HKUST-1). The as-prepared octahedral Mo_xC obtained an overpotential of 151 mV in 1 M KOH (–10 mA cm⁻²). In addition, the construction of heterostructures by doping additional of other metals improved the HER performance of Mo_xC. Ma et al. [17] prepared nitrogen-doped carbon-coated Co-Mo₂C porous heterostructures by the carbonization of a Co-Zn bimetal MOF. Specifically, Co doping provided additional electrons that facilitated charge transfer, which weakened the Mo-H bond strength via an electron synergistic

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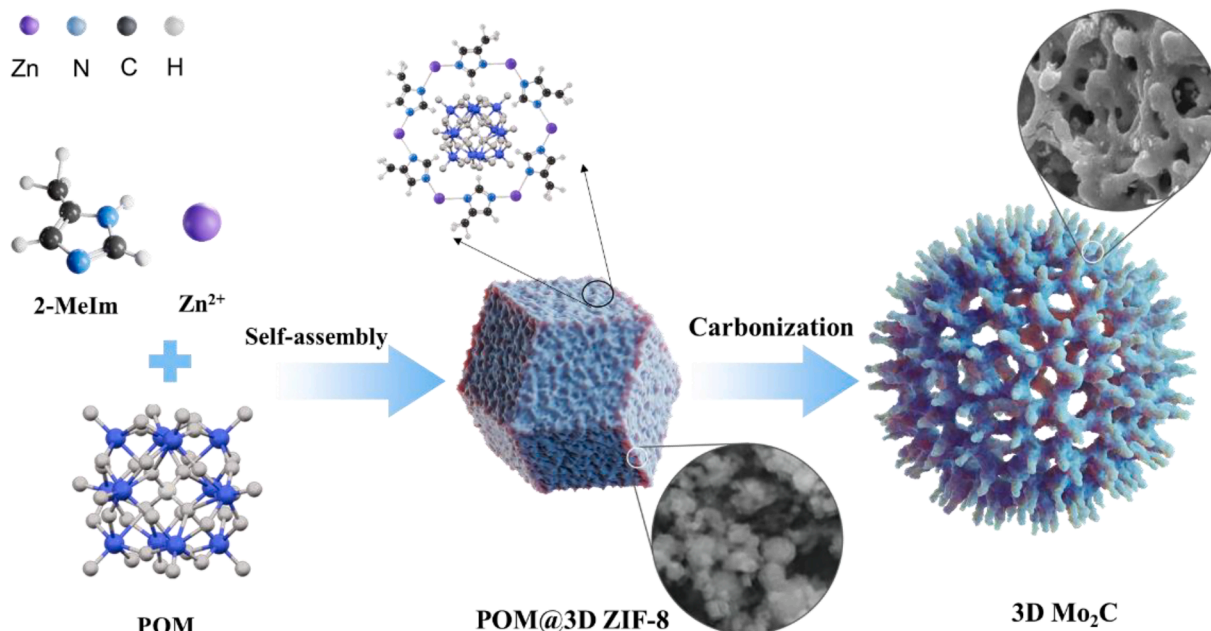


Fig. 1. Synthesis procedure of porous 3D Mo_2C electrocatalyst.

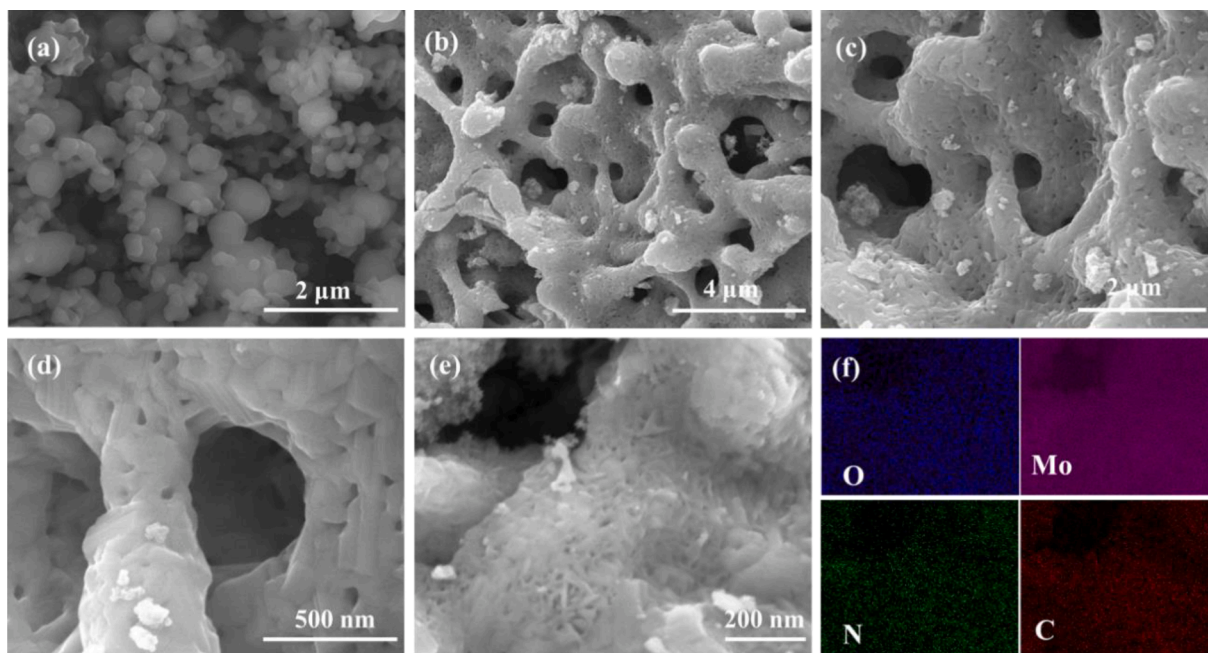


Fig. 2. SEM images of (a) POM@3D ZIF-8, and (b–d) 3D Mo_2C (1:1) electrocatalyst at various magnifications. (e–f) EDS elemental mapping images of 3D Mo_2C (1:1) electrocatalyst.

effect. This promoted the desorption of H^* and reduced the HER overpotential of Mo_2C . Most reported Mo_xC nanoparticles have polycrystalline structures, and the existence of polycrystalline grain boundaries greatly hinders electron transfer [18]. Lin et al. [19] also reported a MoS_2 core blanketed by a single-metal-atom-coordinated, N-doped carbon shell (i.e., $\text{MoS}_2@\text{M}-\text{N}-\text{C}$; $\text{M} = \text{Fe}, \text{Co}, \text{Ni}$) that function as stable ORR/OER bifunctional electrocatalysts. Furthermore, the nitrogen-doped carbon matrix could firmly stabilize the highly energetic single atoms through the metal-nitrogen interaction to mitigate the aggregation of metal atoms [20,21]. In addition, the HER catalytic activity of Mo_xC in different crystal phases is different due to their Fermi level difference. For instance, Leonard et al. [22] reported the HER catalytic

performance of four Mo_xC crystalline phases (α - MoC , β - Mo_2C , γ - MoC , and η - MoC). The results revealed that β - Mo_2C and γ - MoC exhibited the best HER activity. Although a few articles have investigated the effect of different Mo_xC phases on the electrocatalytic activity, the effect of the active crystal face of the electrocatalyst has not been reported.

Herein, a novel coral-like porous nitrogen-doped carbon Mo_2C electrocatalyst (denoted 3D Mo_2C) was synthesized by calcining a POM-doped zeolitic imidazolate framework (ZIF) at 950 °C. The coral-like porous structure exhibited a large pore size, which exposed sufficient active surface sites and facilitated electron transfer to accelerate water electrolysis. The obtained electrocatalyst represented excellent HER catalytic activity in alkaline solution with a low overpotential (110 mV)

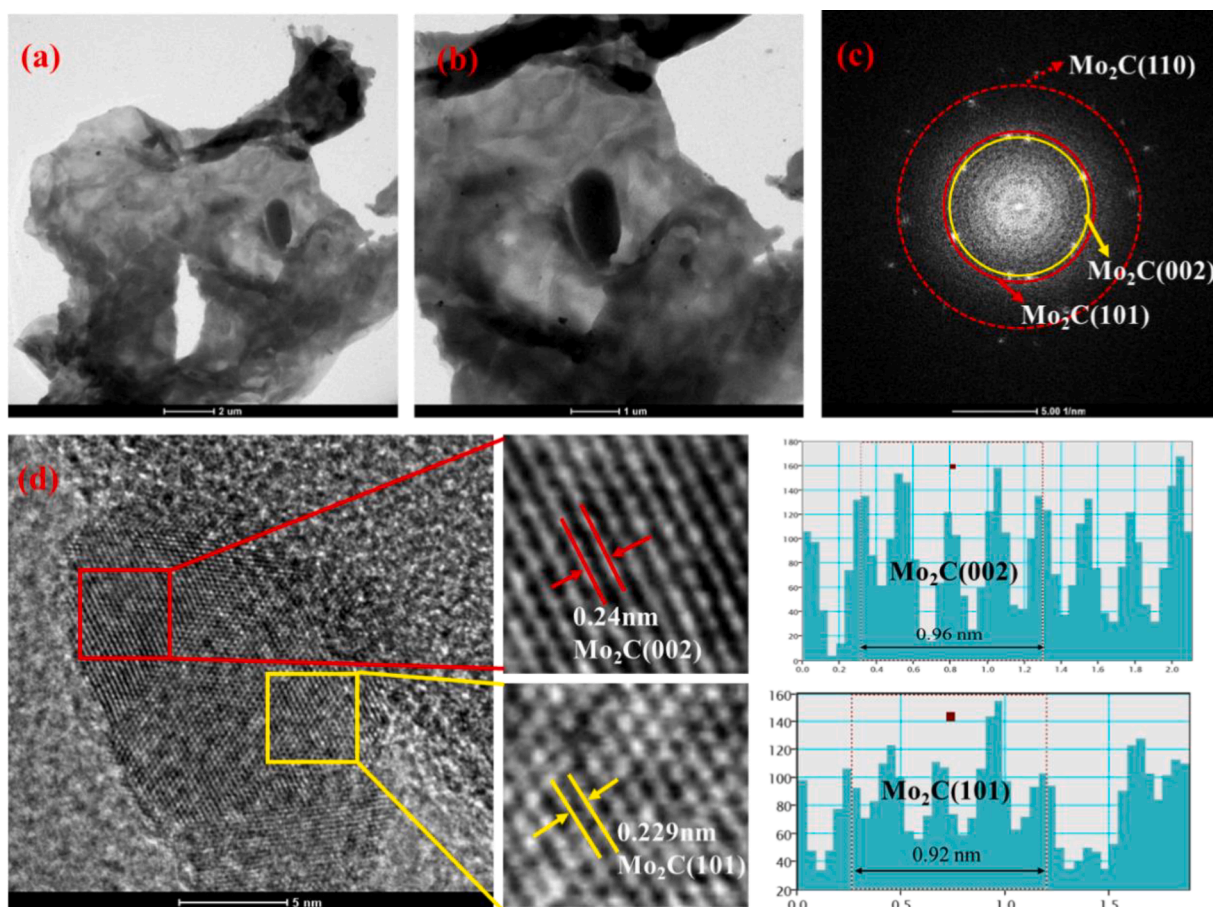


Fig. 3. (a, b) TEM images at various magnifications, (c) SAED pattern, and (d) HRTEM image of 3D Mo₂C (1:1) electrocatalyst.

and small Tafel slope (73.9 mV dec⁻¹). Furthermore, 3D Mo₂C exhibited outstanding stability for 24 h without an obvious overpotential decay in 1.0 M KOH. DFT calculations showed that the moderate (0 0 2)/(1 0 1) crystal face ratio of Mo₂C optimized the adsorption of intermediate OH*, which accelerated the reaction kinetics of the Volmer process and accelerated water dissociation in an alkaline medium. This work provides a novel method for the synthesis of high-efficiency Mo_xC electrocatalysts for hydrogen evolution by active crystal facet regulation.

2. Experimental methods

2.1. Materials

2-Methylimidazole (2-MeIm, >99 %), zinc(II) nitrate hexahydrate (Zn(NO₃)₂·6H₂O, >99 %), phosphomolybdic acid hydrate (POM, >99 %), and polyvinylpyrrolidone (PVP, average mol wt. ~55,000 g/mol) were purchased from Sigma-Aldrich Chemical Reagent Co. Methanol (CH₃OH, >99.99 %), potassium hydroxide (KOH, 95 %), and 20 % Pt/C were obtained from Sinopharm. Nafion solution (5 wt%) was purchased from DuPont Company. Deionized water was obtained using Millipore's Autopure system. All chemicals were used as received without any further purification. Ni foam (NF, 0.5 mm thick) was purchased from Kunshan Tengerhui Electronic Technology Co., Ltd.

2.2. Preparation of electrocatalysts

2.2.1. Preparation of a ZIF-8, POM@ 2D ZIF-8, and POM@ 3D ZIF-8

POM@ 3D ZIF-8 precursor was synthesized by continuous stirring at room temperature. First, 9.85 g (120 mmol) of 2-MeIm was dissolved in 180 mL of methanol and stirred continuously for 30 min to obtain

solution A. Then, 2.947 g Zn(NO₃)₂·6H₂O (10 mmol Zn²⁺) and 0.76 g POM (5 mmol Mo⁶⁺) were dissolved in another 180 mL methanol and then sonicated for 10 min. Then, 1.0 g PVP was added and vigorously stirred for 30 min, denoted as solution B. Finally, solution B was rapidly dropped into solution A, and the mixture was continuously stirred at room temperature (25 °C) for 24 h. After the reaction, the mixture was centrifuged with methanol and ethanol several times. Then, the precipitate was dried in a vacuum drying oven at 60 °C. After drying, the sample was obtained (denoted as POM@ 3D ZIF-8-1:2). POM@ 3D ZIF-8-1:1 and POM@ 3D ZIF-8-2:1 were synthesized by the same method, except the molar ratio of Mo:Zn was adjusted to 1:1 and 2:1.

The synthesis process of ZIF-8 was similar to that of POM@ 3D ZIF-8, except POM was not added, and the molar ratio of Zn(NO₃)₂·6H₂O and 2-MeIm was 1: 8. POM@ 2D ZIF-8 was prepared in the same way as POM@ 3D ZIF-8, except the solvent was changed to deionized water.

2.2.2. Preparation of 3D NC, 2D MoC, and 3D Mo₂C

For the synthesis of 3D Mo₂C-x (x = 2:1, 1:1, 1:2) catalyst, 1.5 g of the dried sample POM@3D ZIF-8-x (x = 2:1, 1:1, 1:2) was placed into a small corundum crucible. Then, the crucible was transferred to a tube furnace, and the sample was calcined at 950 °C (5 °C/min) for 2 h under an atmosphere of high-purity Ar. After calcination, 3D Mo₂C (2:1), 3D Mo₂C (1:1), and 3D Mo₂C (1:2) were obtained as black powders.

For comparison, 2D MoC and 3D NC were calcined under the same conditions, but with different precursors, POM@ 2D ZIF-8 and ZIF-8, respectively.

2.2.3. Preparation of working electrode

First, nickel foam was cut into 1 × 1.5 cm² pieces. Then, the NF pieces were sonicated in ethanol for 15 min to remove surface impurities

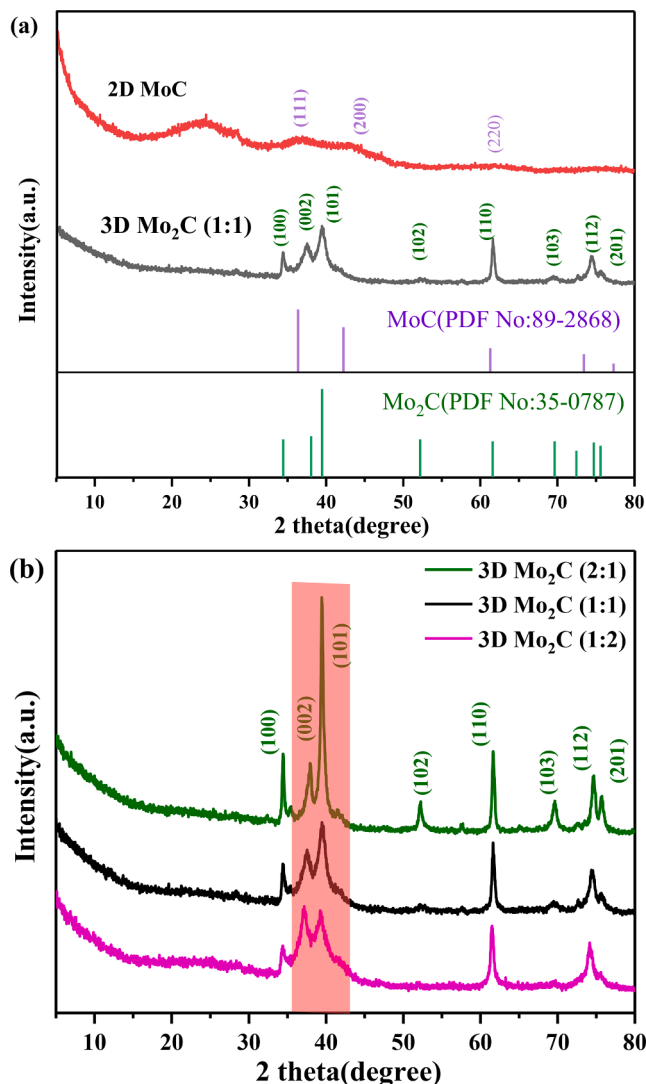


Fig. 4. (a, b) XRD patterns of electrocatalysts: 2D MoC and 3D Mo₂C-*x* (*x* = 2:1, 1:1, 1:2 with various molar ratio of Mo/Zn).

and rinsed with a large amount of deionized water. The pretreated NF pieces were naturally dried for further use. The catalyst ink was prepared as follows: 2 mg of catalyst and 20 μ L of Nafion solution (5 %) were dispersed in 1 mL ethanol solution. Then, the solution was sonicated for 60 min. After the catalyst powder was uniformly dispersed, the catalyst ink was successfully prepared. Finally, 200 μ L of ink was evenly sprayed onto the 1×1.5 cm² NF pieces. The catalyst-loaded NF was used as the working electrode after drying for 30 min under infrared light.

2.3. Characterization

Field emission scanning electron microscopy (FESEM, FEI Scios 2 HiVac) and high-resolution transmission electron microscopy (HRTEM, FEI Tecnai G2 F30.) were used to characterize the micromorphology and surface structure of the samples. Energy-dispersive X-ray spectroscopy (EDS) in the SEM was used to observe the elemental distribution on the catalyst surface. The crystal phase was analyzed by X-ray diffraction (XRD; X'Pert PRO MPD diffractometer) with Cu K α radiation. X-ray photoelectron spectroscopy (XPS; Thermo Scientific) using Al K α radiation was adopted to analyze the chemical states on the catalyst surface. The specific surface areas of the catalysts were determined by the BET method using a Micromeritics Tristar II3020 M instrument. The functional groups on the surface of samples were measured by Fourier-

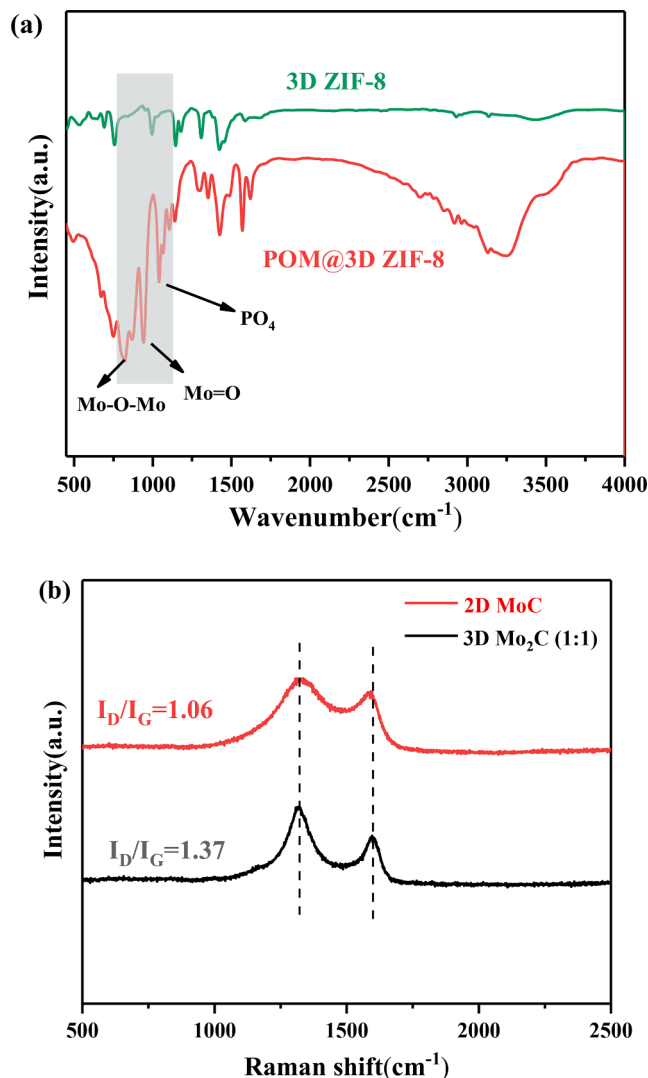


Fig. 5. (a) FTIR spectra of 3D ZIF-8 and POM@3D ZIF-8. (b) Raman spectra of 2D MoC and 3D Mo₂C (1:1).

transform infrared spectrometry (FT-IR, Thermo Scientific Nicolet 10). Raman spectroscopy (Horiba HR800 spectrometer) was used to determine the graphitization degree of catalysts. The contact angle of the electrocatalyst on the electrode surface was investigated using a JY-82B Kruss DSA. Furthermore, we would like to thank the Shiyanjia lab for various characterizations (<https://www.shiyanjia.com>).

2.4. Calculation methods

We used DFT as implemented in the Vienna Ab initio simulation package (VASP) for all calculations [23]. The exchange–correlation potential was described by using the generalized gradient approximation of Perdew–Burke–Ernzerhof (GGA-PBE) [24]. The projector augmented-wave (PAW) method was employed to treat interactions between ion cores and valence electrons [25]. The plane-wave cutoff energy was fixed at 500 eV. Given structural models were relaxed until the Hellmann–Feynman forces were smaller than -0.02 eV/ $\text{\AA}$ and the change in energy was less than 10^{-5} eV. During relaxation, the Brillouin zone was represented by a Γ centered k-point grid of $3 \times 3 \times 1$ for Mo₂C (0 0 2) and MoC and $3 \times 4 \times 1$ for Mo₂C (1 0 1). Spin-polarization was considered in all calculations. To avoid periodic image interactions between two nearest neighbor unit cells, the vacuum was set to 20 $\text{\AA}$ in the z-direction.

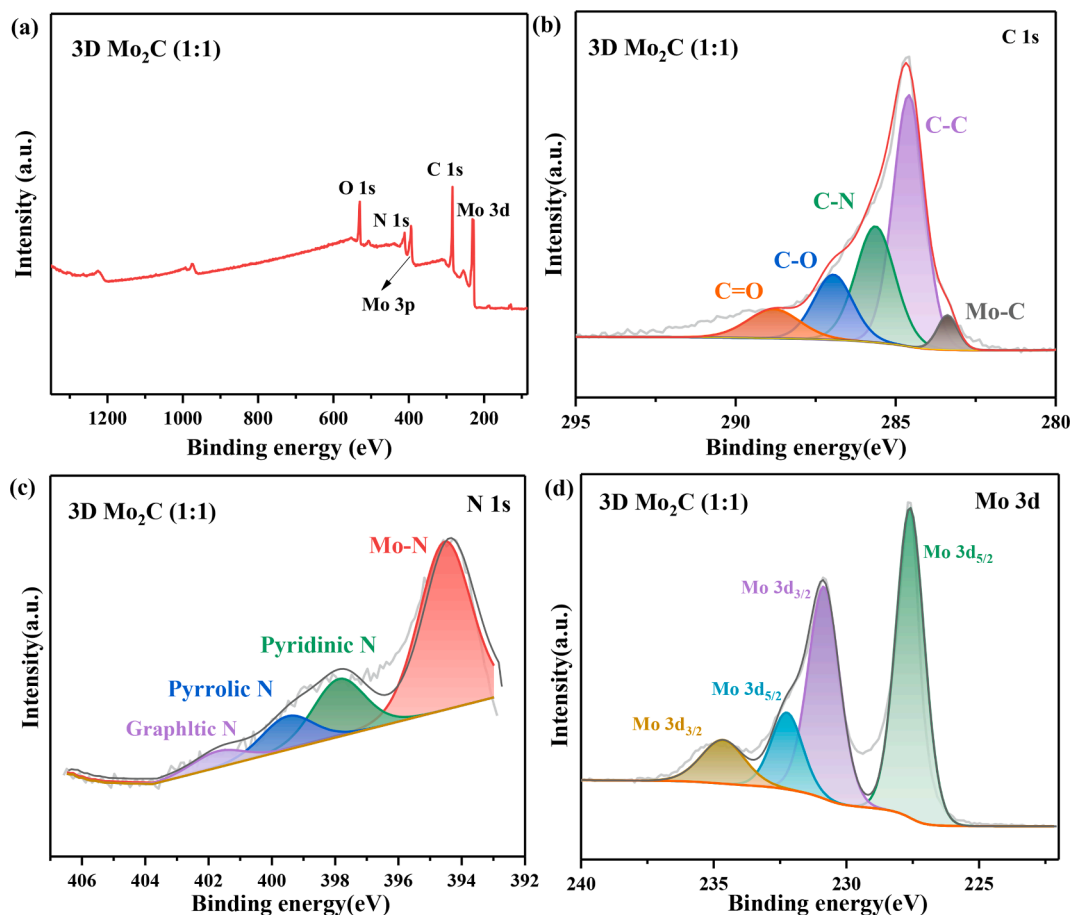


Fig. 6. (a) XPS spectra, and (b) C 1s, (c) N 1s, (d) Mo 3d XPS high-resolution spectra of 3D Mo₂C (1:1).

The free energy of a gas phase molecule or an adsorbate on the surface was calculated by the equation

$$G = E + ZPE - TS$$

where E is the total energy, ZPE is the zero-point energy, T is the temperature in kelvin (298.15 K is used here), and S is the entropy.

2.5. Electrochemical measurements

Electrocatalytic HER tests were carried out using a standard three-electrode system on an electrochemical workstation (Ivium, Netherlands). A graphite rod was used as the counter electrode, Hg/HgO was adopted as the reference electrode, and a piece $1 \times 1.5 \text{ cm}^2$ NF-supported catalyst was used as the working electrode. Before the test, the solution was continuously purged with high-purity nitrogen gas for 30 min to remove oxygen. The polarization curves of the catalysts were tested in 1 M KOH (pH = 13.5) solution with a scan rate of 5 mV/s (polarization curve test without IR calibration). The Tafel plots were converted from the polarization curves. Electrochemical impedance spectroscopy (EIS) was performed in the frequency range between 10^5 and 10^{-1} Hz to analyze the charge transfer resistance of the catalysts. Cyclic voltammetry (CV) was performed with six different scanning speeds in the non-Faradaic region (20, 40, 60, 80, 100, and 120 mV/s). The electrical double layer capacitance (C_{dl}) was calculated from the CV curves and was used to evaluate the electrochemical active surface area (ECSA). For stability tests, cyclic voltammetry and chronopotentiometry measurements were performed to determine the performance of catalysts. All potentials were converted to reversible hydrogen electrode (RHE) voltage ($E_{RHE} = E_{Hg/HgO} + 0.098 + 0.059 \times \text{pH}$).

3. Results and discussion

3.1. Characterization of the catalysts

Fig. 1 schematically illustrates the synthesis process of the porous coral-like Mo₂C catalyst. First, the precursor POM@3D ZIF-8 was synthesized via *in-situ* doping of POM into the ZIF-8 framework. Then, the prepared precursor POM@3D ZIF-8 was calcined at 950 °C for 2 h. A special coral-like porous Mo₂C catalyst was formed due to the confinement of ZIF-8 avoiding the aggregation of POM during calcination.

As shown in Figs. 2 and S1, the morphological characteristics of the catalysts were analyzed by SEM. In Fig. 2(a), the morphology of the prepared POM@3D ZIF-8 precursor was not a regular dodecahedron (similar to that of the original ZIF-8). Instead, many nanoparticles with different sizes aggregated on the surface. The doped POM might have affected the surface morphology [26]. Fig. 2(b–e) exhibit the SEM morphologies of 3D Mo₂C (1:1) at different scales. After high-temperature calcination, the surface of 3D Mo₂C (1:1) became rough and formed abundant mesoporous tunnels. Compared with the formicary-like morphology of the precursor, the sample showed a porous coral-like structure, possibly due to the volatilization of Zn ions [27]. The morphologies of the control samples are presented in Fig. S1, including POM@2D ZIF-8, 2D MoC, 3D Mo₂C (2:1), and 3D Mo₂C (1:2) catalysts. The POM@2D ZIF-8 and 2D MoC samples exhibited regular leaf-like morphologies. 3D Mo₂C (1:1) retained its original dodecahedral morphology with slight agglomeration, indicating that too little POM was added to form a porous structure. The structure of 3D Mo₂C (1:2) was similar to that of 3D Mo₂C (1:1), but the pore size was smaller due to severe surface nanoparticle aggregation. The EDS elemental mapping of Fig. 2(f) also illustrates the uniform distribution of Mo, C, N, and O.

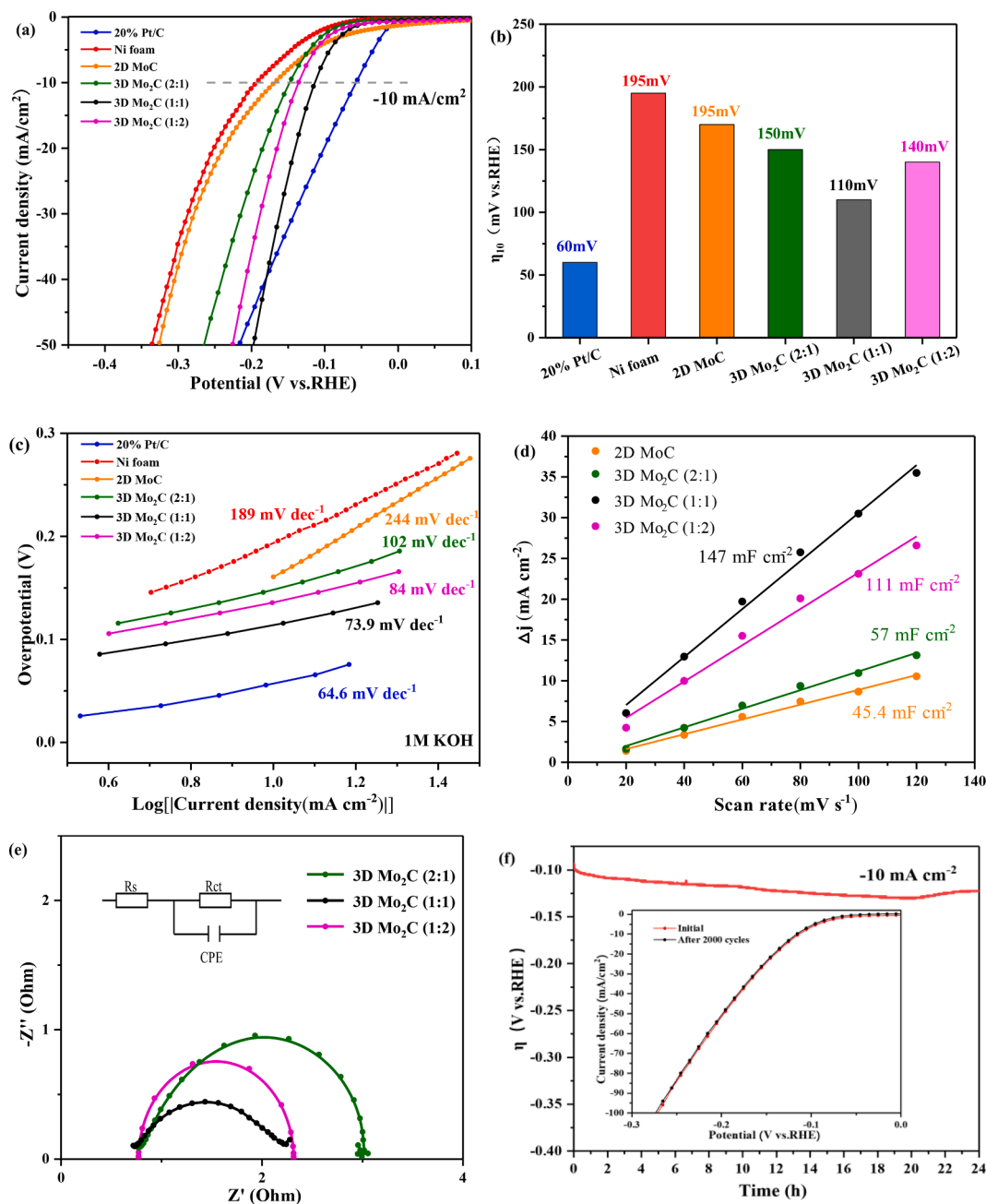


Fig. 7. (a) Polarization curves, (b) HER activity (overpotentials at $10 \text{ mA}/\text{cm}^2$) and (c) Tafel plots of 2D MoC, 3D Mo₂C- x ($x = 2:1, 1:1, 1:2$), Ni foam and 20 % Pt/C. (d) Estimated C_{dl} and relative ECSA of 2D MoC and 3D Mo₂C- x ($x = 2:1, 1:1, 1:2$). (e) Nyquist plots of 3D Mo₂C- x ($x = 2:1, 1:1, 1:2$). (f) Stability test after 2000 CV cycles and chronopotentiometry test of 3D Mo₂C (1:1).

The morphology and surface structure of the catalysts were further characterized by TEM. As shown in Fig. 3(a, b), the morphology of 3D Mo₂C (1:1) was irregular and porous, which was consistent with the SEM results. The special porous channel facilitated the transfer of electrons and hydrogen desorption, which promoted the alkaline HER kinetics. As shown in Fig. 3(c, d), the selected area electron diffraction (SAED) and high-resolution TEM (HRTEM) images were used to analyze the crystal planes of 3D Mo₂C (1:1). The HRTEM images and corresponding fast Fourier transform (FFT) images clearly showed two distinct lattice spacings of 0.24 and 0.229 nm, corresponding to the (0 0 2) and (1 0 1) crystal planes of Mo₂C. The polycrystalline ring appearing in the SAED proved the existence of different Mo₂C crystal planes. The TEM results demonstrated the excellent crystallinity of the porous Mo₂C catalyst, and the presence of the ZIF framework prevented nanoparticle

aggregation.

The X-ray diffraction (XRD) patterns were obtained to further characterize the phase composition of 2D MoC and 3D Mo₂C- x ($x = 2:1, 1:1, 1:2$), as shown in Fig. 4(a, b). The weak peak at 21.5° corresponded to amorphous carbon. 2D MoC exhibited three distinct diffraction peaks at 36.4° , 42.3° , and 61.4° , which corresponded to the (1 1 0), (2 0 0), and (2 2 0) crystal planes (JCPDS No. 89-2868), respectively. The diffraction peaks of 3D Mo₂C- x ($x = 2:1, 1:1, 1:2$) corresponded to the standard card JCPDS No. 35-0787, which confirmed the existence of β -Mo₂C. In Fig. 4(b), 3D Mo₂C- x exhibited similar crystal phases at different Zn: Mo ratios. As the amount of POM gradually increased, the (1 0 1) diffraction peak gradually weakened, and the (0 0 2) diffraction peak increased significantly. These XRD results indicate that the POM doping content could be used to tune the relative content ratio of Mo₂C crystal facets.

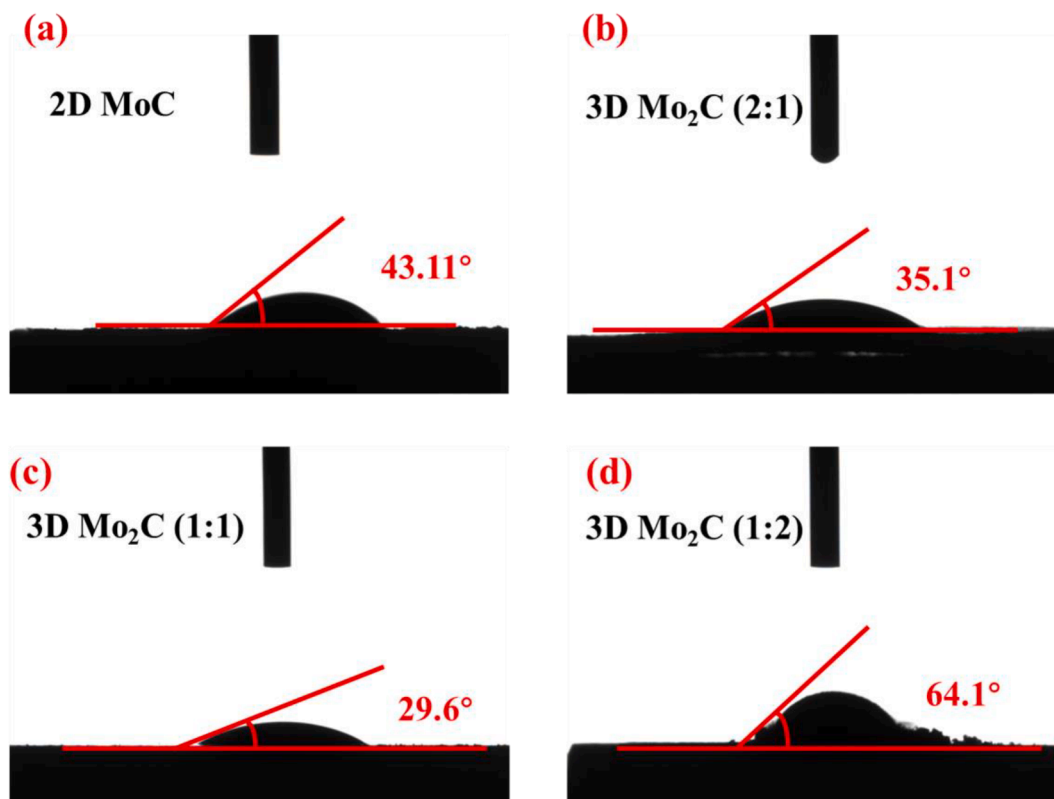


Fig. 8. (a–d) Contact angles of 2D MoC and 3D Mo₂C-*x* (*x* = 2:1, 1:1, 1:2).

This might be because Zn sublimated to form pores at high temperatures, and regulating the proportion of Zn can control the growth of Mo₂C crystal planes. In addition, XRD semi-quantitative calculations were used to analyze the relative content ratio of Mo₂C (0 0 2) and (1 0 1) crystal planes as shown in Table S1. 3D Mo₂C (1:1) exhibited a moderate facet ratio (0.766) relative to 3D Mo₂C (2:1) (0.241) and 3D Mo₂C (1:2) (2.698) catalysts. The results indicated that (0 0 2) and (1 0 1) crystal planes contributed to synergistic catalytic effect for HER.

Fig. 5(a) exhibits the FT-IR spectra of 3D ZIF-8 and POM@ 3D ZIF-8, which confirmed that the POM was successfully doped into the ZIF-8 structure. Compared with the FT-IR spectrum of 3D ZIF-8, the POM@ 3D ZIF-8 spectrum exhibited an additional strong peak near 812 cm⁻¹, corresponding to the Mo–O–Mo bond. In addition, the peaks at 935 cm⁻¹ and 1037 cm⁻¹ were assigned to Mo=O and PO₄, respectively [28]. The peak at 3500 cm⁻¹ might be due to bonded water in the sample. The Raman spectra of 2D MoC and 3D Mo₂C (1:1) were analyzed, as shown in Fig. 5(b). Generally, the D-band (~1350 cm⁻¹) and G-band (~1580 cm⁻¹) reflect the defects and disorder in the crystal structure and the vibrations of sp² carbons [29]. A larger I_D/I_G value indicates more carbon defects in catalysts. The I_D/I_G value of 2D MoC and 3D Mo₂C (1:1) reached 1.06 and 1.37, respectively. The results showed that 3D Mo₂C (1:1) contained more carbon defects, which may expose more Mo active sites and accelerate the dissociation of water.

The Brunauer-Emmett-Teller (BET) method was used for surface area analysis, and the corresponding N₂ adsorption–desorption isotherms are shown in Fig. S3. The BET specific surface areas of the original POM@ 2D ZIF-8, POM@ 3D ZIF-8, 2D MoC, and 3D Mo₂C (1:1) samples were 97.26, 37.25, 515.74, and 81.04 m²/g, respectively. After calcination, the BET surface area of both MoC and Mo₂C increased compared with that of the precursor. Moreover, the average pore diameters of POM@ 2D ZIF-8, POM@ 3D ZIF-8, 2D MoC, and 3D Mo₂C (1:1) were 2.49, 3.83, 2.65, and 22.70 nm, respectively (Table S2). This indicated that a higher degree of mesoporosity was achieved after calcination. Micropores can provide larger specific surface area and pore volume and increase the

number of active sites, mesopores and macropores can promote the rapid diffusion of ions and the rapid transfer of substances. 3D Mo₂C (1:1) showed the largest average pore size, which facilitated the contact between the electrode surface and the electrolyte and enhanced charge and mass transfer.

The surface composition and chemical states of porous 3D Mo₂C were analyzed by X-ray photoelectron spectroscopy (XPS). As shown in Fig. 6(a), the XPS survey spectrum revealed the presence of C, N, Mo, and O elements in 3D-Mo₂C. The high-resolution C 1s, N 1s, and Mo 3d XPS spectra of 3D Mo₂C (1:1) are displayed in Fig. 6(b–d). The C 1s spectrum was divided into five peaks. The four main peaks at 284.7, 285.6, 287.2, and 288.8 eV correspond to C–C, C–N, C–O, and C=O, respectively [30,31]. There was also a small peak at 283.4 eV, which may be the result of Mo–C coordination in Mo₂C [32]. As shown in Figs. S4, 2D MoC also exhibited four main peaks corresponding to C–C, C–N, C–O, and C=O. However, there was no small Mo–C peak, which may indicate the poor crystallinity of 2D MoC. The high-resolution N 1s spectrum was divided into four peaks at binding energies of 394.58, 398.0, 399.6, and 401.3 eV, which were related to Mo–N, pyridinic N, pyrrolic N, and graphitic N, respectively [33]. The results also indicated that the product obtained after carbonization was a composite of Mo₂C particles embedded in N-doped carbon. The high-resolution XPS Mo 3d spectrum of 3D Mo₂C (1:1) is depicted in Fig. 6(d), which was divided into four peaks. The peaks at binding energies of 227.58 eV and 230.9 eV corresponded to Mo²⁺ of Mo₂C [34]. Other peaks were observed at 232.28 eV (Mo⁴⁺ of MoO₂) and 234.88 eV (Mo⁶⁺ of MoO₃) [35]. These oxides might have been formed by the oxidation of the Mo₂C surface exposed to the air. DFT calculations demonstrated that Mo²⁺ served as the main active site for the alkaline HER. The additional oxygen atoms of MoO₂ and MoO₃ oxides facilitated the H₂O* adsorption and accelerated the reaction kinetics [36]. However, excess oxide hindered the exposure of Mo²⁺ active sites [37]. According to Tables S3 and S4, 3D Mo₂C (1:1) contained more Mo²⁺ active sites than 2D MoC, which exposed more active sites for the alkaline HER.

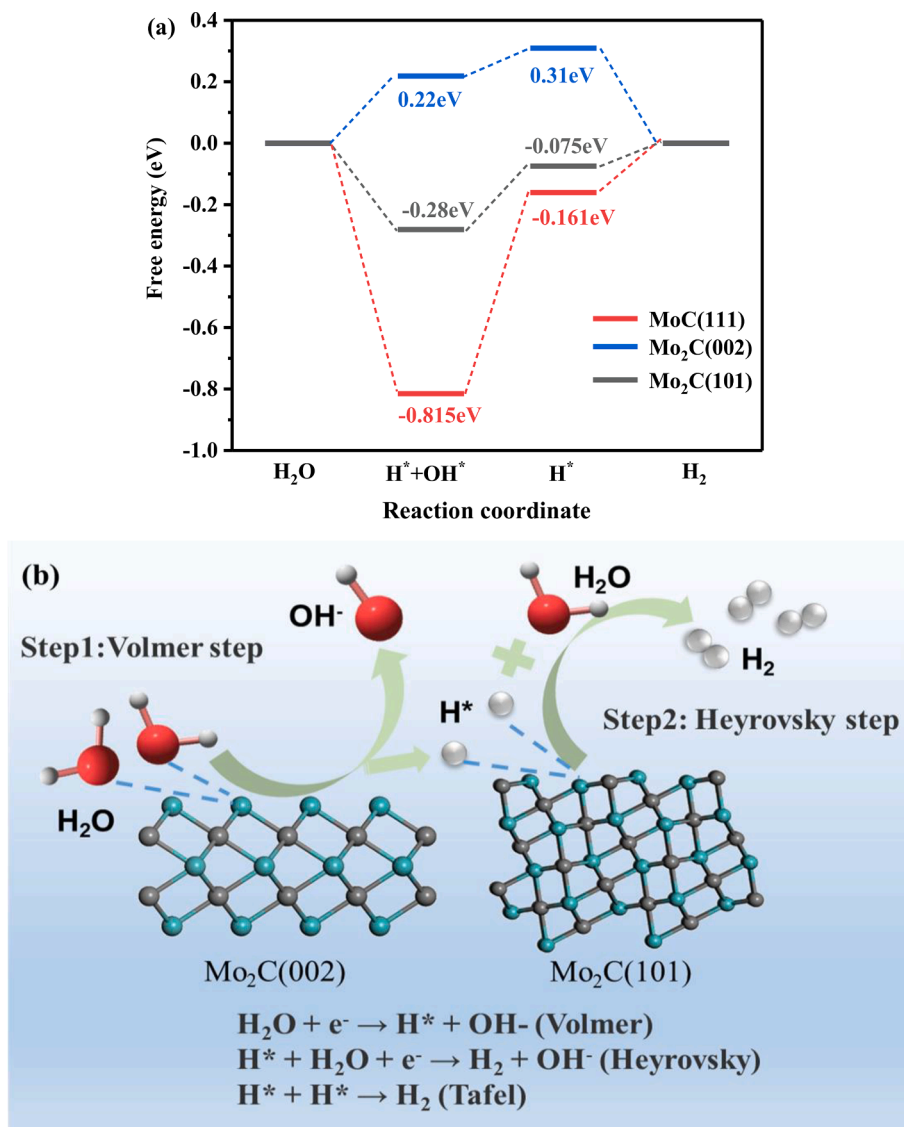


Fig. 9. DFT calculations. (a) Hydrogen absorption Gibbs free energy for HER on 2D MoC (1 1 1), 3D Mo₂C (0 0 2) and 3D Mo₂C (1 0 1). (b) Schematic of reaction pathways of HER electrocatalysis on 3D Mo₂C catalyst.

3.2. Catalytic activity for the HER

The HER performance of the catalysts was evaluated by LSV in 1 M KOH. None of the electrochemical tests was iR-corrected. As shown in Fig. 7(a, b), the performance of 2D MoC, 3D Mo₂C (2:1), 3D Mo₂C (1:1), 3D Mo₂C (1:2), NF, and commercial Pt/C were tested under identical conditions. The 3D Mo₂C (1:1) catalyst exhibited a low overpotential ($\eta_{10} = 110$ mV) and outperformed other electrocatalysts, such as 2D MoC (195 mV), 3D Mo₂C (2:1) (150 mV), 3D Mo₂C (1:2) (140 mV), and NF (195 mV). The coral-like porous structure facilitated rapid charge and reactant transport, which was the main reason for the catalyst's excellent performance.

The reaction kinetics of the catalysts were characterized by the Tafel slopes, as shown in Fig. 7(c). 3D Mo₂C (1:1) exhibited a lower Tafel slope (73.9 mV dec^{-1}) than 2D MoC (244 mV dec^{-1}), 3D Mo₂C (2:1) (102 mV dec^{-1}), 3D Mo₂C (1:2) (84 mV dec^{-1}), and NF (189 mV dec^{-1}) and was the closest to that of 20 % Pt/C (64.6 mV dec^{-1}). The results show that 3D Mo₂C had the fastest alkaline HER kinetics.

The ECSA values of the catalysts were calculated from the CV curves at different scanning rates based on the double-layer capacitance (C_{dl}) theory. As shown in Figs. S5 and 7(d), 3D Mo₂C (1:1) exhibited a C_{dl} of

147 mF cm^{-2} , which was higher than that of 2D MoC (45.4 mF cm^{-2}), 3D Mo₂C (2:1) (57 mF cm^{-2}), and 3D Mo₂C (1:2) (111 mF cm^{-2}). This result indicated that 3D Mo₂C (1:1) had a larger electrocatalytic effective area and exposed more Mo active sites, which promoted the HER performance. This was due to the special porous coral-like composite structure of 3D Mo₂C, which adsorbed more H* and produced H₂.

The charge transfer resistance (R_{ct}) of the HER was analyzed by electrochemical impedance spectroscopy (EIS). As shown in Fig. 7(e), according to the equivalent circuit diagram, the R_{ct} depended on the diameter of the semicircle in the Nyquist plot. 3D Mo₂C (1:1) exhibited a smaller R_{ct} (1.4Ω) than 3D Mo₂C (1:2) (1.55Ω) and 3D Mo₂C (2:1) (2.2Ω). The results indicated that 3D Mo₂C (1:1) exhibited an excellent electron transfer rate and accelerated the electrocatalytic HER.

The stability of the catalysts was evaluated using long-term chronopotentiometry and CV. As shown in Fig. 7(f), the overpotential of decayed only from 110 mV to 122 mV during continuous HER for 24 h. The polarization curve of 3D Mo₂C (1:1) nearly overlapped before and after 2000 CV scans. 3D Mo₂C (1:1) exhibited excellent stability during the alkaline HER. In the 3D Mo₂C (1:1) composite structure, the Mo₂C nanoparticles were encapsulated by nitrogen-doped carbon, which prevented their mutual aggregation and ensured their strong adhesion

on the carbon surface. The coral-like porous structure provided sufficient surface area that exposed more Mo sites and enhanced the interfacial electronic interactions. These promoted the long-term stability of the catalyst during the HER [38].

3.3. Theoretical calculations and catalytic mechanism

The Tafel slope can be used to determine the mechanism of the electrocatalytic HER. 3D Mo₂C (1:1) exhibited a Tafel slope of 73.9 mV dec⁻¹, indicating that the HER on the catalyst surface may be based on the Volmer-Heyrovsky mechanism [39]. In alkaline processes, the Volmer process ($\text{H}_2\text{O} + \text{e}^- \rightarrow \text{H}^* + \text{OH}^-$) is the rate-limiting step. Therefore, strong adsorption and rapid dissociation of water on the catalyst surface were crucial for the subsequent Heyrovsky step ($\text{H}^* + \text{H}_2\text{O} + \text{e}^- \rightarrow \text{H}_2 + \text{OH}^-$) [40]. According to the XPS analysis, the surface of 3D Mo₂C (1:1) contained MoO₂ and MoO₃ oxygen-containing groups, which can adsorb water molecules. As shown in Fig. 8, the hydrophilicity of the catalyst was demonstrated by its water contact angle (CA). 3D Mo₂C (1:1) possessed the best hydrophilicity with a water contact angle of 29.6°, which was smaller than that of the other samples: 2D MoC (43.11°), 3D Mo₂C (2:1) (35.1°), and 3D Mo₂C (2:1) (64.1°). In addition, the 3D Mo₂C coral-like structure had many mesopores and channels, which facilitated the penetration of water molecules during the HER reaction.

DFT calculations were performed to further understand the enhanced HER performance of 3D Mo₂C. The Gibbs free energy of absorption for the HER intermediates OH* and H* on MoC (1 1 1), Mo₂C (0 0 2), and Mo₂C (1 0 1) were calculated, as shown in Fig. 9(a). The free energy for water dissociation (ΔG_{OH^*}) on MoC (1 1 1), Mo₂C (0 0 2), and Mo₂C (1 0 1) were -0.815, 0.22, and -0.28 eV, respectively. They exhibited a smaller adsorption energy than 2D MoC, indicating that 3D Mo₂C was more conducive to the dissociation of water and had better HER activity. Water dissociation (the Volmer process) is the rate-limiting step for the alkaline HER, which explains the poor performance of 2D MoC. We also found that the adsorption energies of OH* and H* were different for different crystal faces of Mo₂C. For Mo₂C (0 0 2) and Mo₂C (1 0 1), the ΔG_{OH^*} were calculated to be 0.22 eV and -0.28 eV, respectively, indicating that Mo₂C (0 0 2) was more likely to undergo water dissociation to form OH* and H* (Volmer process). For the hydrogen absorption Gibbs free energy (ΔG_{H^*}), Mo₂C (1 0 1) exhibited smaller ΔG_{H^*} (-0.075 eV) than Mo₂C (0 0 2) (0.31 eV). The ΔG_{H^*} of Mo₂C (1 0 1) nearly reached thermal neutrality ($\Delta G_{\text{H}^*} = 0$), indicating that H* adsorbed on the Mo₂C (1 0 1) facet was easily desorbed as hydrogen. Thus, the Mo₂C (0 0 2) and (1 0 1) crystal planes had a synergistic effect during the alkaline HER. However, too many exposed Mo₂C (0 0 2) facets inhibited subsequent desorption of H* to generate H₂. Too many exposed Mo₂C (1 0 1) facets hindered the dissociation of water. DFT calculations showed that regulating the exposure ratio of Mo₂C on different crystal surfaces reduced the intermediate adsorption energy barriers for OH* and H* during the hydrogen evolution process and enhanced the alkaline HER performance. Therefore, the alkaline HER mechanism on the Mo₂C surface may be as follows (Fig. 9b): H₂O dissociated on the Mo₂C (0 0 2) surface to form H* and OH*, and then H* was transferred to Mo₂C (1 0 1). Then, H₂ was formed and released from the Mo₂C surface.

4. Conclusion

A porous coral-like Mo₂C catalyst was synthesized by a ZIF-8 derivatization method using a POM as the molybdenum source. The ZIF-8 template played a key role in the formation of coral-like porous Mo₂C. The volatilization of Zn ions during high-temperature calcination increased the electrochemically active specific surface area and exposed more Mo active sites. The prepared 3D Mo₂C (1:1) catalyst achieved a low overpotential of 110 mV (η_{10}) with a Tafel slope of 73.9 mV dec⁻¹ in the alkaline HER process. XRD characterization proved that the

exposure ratio of Mo₂C (0 0 2) and (1 0 1) crystal planes could be regulated by changing the doping amount of POM. The Mo₂C (1 0 1) and (0 0 2) crystal planes played a synergistic role during the alkaline HER process. The DFT calculations indicated that the Mo₂C (0 0 2) facet favored water dissociation, while the Mo₂C (1 0 1) facet was more likely to adsorb H* and desorb H₂. Therefore, 3D Mo₂C with a suitable (1 0 1): (0 0 2) crystal plane ratio showed the best alkaline HER performance. In short, this work provides new possibilities for the crystal facet regulation of Mo_xC catalysts.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cej.2022.138977>.

References

- [1] Y. Yang, H. Yao, Z. Yu, S.M. Islam, H. He, M. Yuan, Y. Yue, K. Xu, W. Hao, G. Sun, H. Li, S. Ma, P. Zapol, M.G. Kanatzidis, Hierarchical nanoassembly of MoS₂/Co₉S₈/Ni₃S₂/Ni as a highly efficient electrocatalyst for overall water splitting in a wide pH range, *J. Am. Chem. Soc.* 141 (2019) 10417–10430.
- [2] H. Wu, H.N. Alshareef, T. Zhu, Photo-assisted electrochemical hydrogen evolution by plasmonic Ag nanoparticle/nanorod heterogeneity, *InfoMat* 1 (2019) 417–425.
- [3] J. Joo, T. Kim, J. Lee, S.I. Choi, K. Lee, Morphology-controlled metal sulfides and phosphides for electrochemical water splitting, *Adv. Mater.* 31 (2019) e1806682.
- [4] Z. Zhao, H. Liu, W. Gao, W. Xue, Z. Liu, J. Huang, X. Pan, Y. Huang, Surface-engineered PtNi-O nanostructure with record-high performance for electrocatalytic hydrogen evolution reaction, *J. Am. Chem. Soc.* 140 (2018) 9046–9050.
- [5] M.A.R. Anjum, J.S. Lee, Sulfur and nitrogen dual-doped molybdenum phosphide nanocrystallites as an active and stable hydrogen evolution reaction electrocatalyst in acidic and alkaline media, *ACS Catal.* 7 (2017) 3030–3038.
- [6] J.O.M. Bockris, E.C. Potter, The mechanism of the cathodic hydrogen evolution reaction, *J. Electrochem. Soc.* 99 (1952) 169.
- [7] Z. Dai, H. Geng, J. Wang, Y. Luo, B. Li, Y. Zong, J. Yang, Y. Guo, Y. Zheng, X. Wang, Q. Yan, Hexagonal-phase cobalt monophosphosulfide for highly efficient overall water splitting, *ACS Nano* 11 (2017) 11031–11040.
- [8] D. Geng, X. Zhao, Z. Chen, W. Sun, W. Fu, J. Chen, W. Liu, W. Zhou, K.P. Loh, Direct synthesis of large-area 2D Mo₂C on in situ grown graphene, *Adv. Mater.* 29 (2017).
- [9] H.B. Wu, B.Y. Xia, L. Yu, X.Y. Yu, X.W. Lou, Porous molybdenum carbide nano-octahedrons synthesized via confined carburization in metal-organic frameworks for efficient hydrogen production, *Nat. Commun.* 6 (2015) 6512.
- [10] H. Sun, X. Xu, Y. Song, W. Zhou, Z. Shao, Designing high-valence metal sites for electrochemical water splitting, *Adv. Funct. Mater.* 31 (2021).
- [11] S. Wang, J. Wang, M. Zhu, X. Bao, B. Xiao, D. Su, H. Li, Y. Wang, Molybdenum-carbide-modified nitrogen-doped carbon vesicle encapsulating nickel nanoparticles: a highly efficient, low-cost catalyst for hydrogen evolution reaction, *J. Am. Chem. Soc.* 137 (2015) 15753–15759.
- [12] G. Liu, K. Wang, L. Wang, B. Wang, Z. Lin, X. Chen, Y. Hua, W. Zhu, H. Li, J. Xia, A Janus cobalt nanoparticles and molybdenum carbide decorated N-doped carbon for high-performance overall water splitting, *J. Colloid Interface Sci.* 583 (2021) 614–625.
- [13] L. Fan, T. Meng, Q. Li, D. Wang, Z. Xing, E. Wang, X. Yang, Ru nanoparticles encapsulated in ZIFs-derived porous N-doped hierarchical carbon nanofibers for enhanced hydrogen evolution reaction, *Catal. Sci. Technol.* 10 (2020) 7302–7308.
- [14] T.V.M. Sreekanth, G.R. Dillip, P.C. Nagajothi, K. Yoo, J. Kim, Integration of marigold 3D flower-like Ni-MOF self-assembled on MWCNTs via microwave irradiation for high-performance electrocatalytic alcohol oxidation and oxygen evolution reactions, *Appl. Catal. B* 285 (2021).

- [15] H.F. Wang, L. Chen, H. Pang, S. Kaskel, Q. Xu, MOF-derived electrocatalysts for oxygen reduction, oxygen evolution and hydrogen evolution reactions, *Chem. Soc. Rev.* 49 (2020) 1414–1448.
- [16] Y.-T. Liao, V.C. Nguyen, N. Ishiguro, A.P. Young, C.-K. Tsung, K.C.W. Wu, Engineering a homogeneous alloy-oxide interface derived from metal-organic frameworks for selective oxidation of 5-hydroxymethylfurfural to 2,5-furandicarboxylic acid, *Appl. Catal. B* 270 (2020).
- [17] P. Zhang, Y. Liu, T. Liang, E.H. Ang, X. Zhang, F. Ma, Z. Dai, Nitrogen-doped carbon wrapped Co-Mo₂C dual mott-schottky nanosheets with large porosity for efficient water electrolysis, *Appl. Catal. B* 284 (2021).
- [18] K. Zhang, G. Zhang, J. Qu, H. Liu, Tungsten-assisted phase tuning of molybdenum carbide for efficient electrocatalytic hydrogen evolution, *ACS Appl. Mater. Interfaces* 10 (2018) 2451–2459.
- [19] Y. Yan, S. Liang, X. Wang, M. Zhang, S.-M. Hao, X. Cui, Z. Li, Z. Lin, Robust wrinkled MoS₂/N-C bifunctional electrocatalysts interfaced with single Fe atoms for wearable zinc-air batteries, *Proc. Natl. Acad. Sci. U.S.A.* 118 (2021).
- [20] M. Liu, P. Zhang, Z. Qu, Y. Yan, C. Lai, T. Liu, S. Zhang, Conductive carbon nanofiber interpenetrated graphene architecture for ultra-stable sodium ion battery, *Nat. Commun.* 10 (2019) 3917.
- [21] Y. Yan, P. Zhang, Z. Qu, M. Tong, S. Zhao, Z. Li, M. Liu, Z. Lin, Carbon/Sulfur aerogel with adequate mesoporous channels as robust polysulfide confinement matrix for highly stable lithium-sulfur battery, *Nano Lett.* 20 (2020) 7662–7669.
- [22] C. Wan, Y.N. Regmi, B.M. Leonard, Multiple phases of molybdenum carbide as electrocatalysts for the hydrogen evolution reaction, *Angew. Chem. Int., Ed. Engl.* 53 (2014) 6407–6410.
- [23] J.P. Perdew, K. Burke, M. Ernzerhof, Generalized gradient approximation made simple, *Phys. Rev. Lett.* 77 (1996) 3865–3868.
- [24] G. Kresse, D. Joubert, From ultrasoft pseudopotentials to the projector augmented-wave method, *Phys. Rev. B* 59 (1999) 1758–1775.
- [25] S.N. Steinmann, C. Corminboeuf, A density dependent dispersion correction, *Chimia (Aarau)* 65 (2011) 240–244.
- [26] X. Zhao, X. He, F. Yin, B. Chen, G. Li, H. Yin, Zeolite imidazolate framework-8 derived molybdenum carbide/nitrogen-doped carbon for highly-efficient hydrogen evolution reaction, *Int. J. Hydrogen Energy* 45 (2020) 15483–15494.
- [27] C. Yang, S. Yun, J. Shi, M. Sun, N. Zafar, A. Arshad, Y. Zhang, L. Zhang, Tailoring the supercapacitive behaviors of Co/Zn-ZIF derived nanoporous carbon via incorporating transition metal species: a hybrid experimental-computational exploration, *Chem. Eng. J.* 419 (2021).
- [28] M. Jafarinasab, A. Akbari, Co-ZIF-67 encapsulated phosphomolybdic acid as a hybrid catalyst for deep oxidative desulfurization, *J. Environ. Chem. Eng.* 9 (2021).
- [29] X. Yang, J. Cheng, X. Yang, Y. Xu, W. Sun, J. Zhou, MOF-derived Cu@Cu₂O heterogeneous electrocatalyst with moderate intermediates adsorption for highly selective reduction of CO₂ to methanol, *Chem. Eng. J.* 431 (2022).
- [30] Y. Ma, T. Yang, H. Zou, W. Zang, Z. Kou, L. Mao, Y. Feng, L. Shen, S.J. Pennycook, L. Duan, X. Li, J. Wang, Synergizing Mo single atoms and Mo₂C nanoparticles on CNTs synchronizes selectivity and activity of electrocatalytic N₂ reduction to ammonia, *Adv. Mater.* 32 (2020) e2002177.
- [31] Y. Chen, Z. Li, Y. Zhu, D. Sun, X. Liu, L. Xu, Y. Tang, Atomic Fe dispersed on N-doped carbon hollow nanospheres for high-efficiency electrocatalytic oxygen reduction, *Adv. Mater.* 31 (2019) e1806312.
- [32] F. Yu, Y. Gao, Z. Lang, Y. Ma, L. Yin, J. Du, H. Tan, Y. Wang, Y. Li, Electrocatalytic performance of ultrasmall Mo₂C affected by different transition metal dopants in hydrogen evolution reaction, *Nanoscale* 10 (2018) 6080–6087.
- [33] Y. Yi, W. Zhao, Z. Zeng, C. Wei, C. Lu, Y. Shao, W. Guo, S. Dou, J. Sun, ZIF-8@ZIF-67-derived nitrogen-doped porous carbon confined CoP polyhedron targeting superior potassium-ion storage, *Small* 16 (2020) e1906566.
- [34] T. Xiong, J. Jia, Z. Wei, L. Zeng, Y. Deng, W. Zhou, S. Chen, N-doped carbon-wrapped MoC heterophase sheets for high-efficiency electrochemical hydrogen production, *Chem. Eng. J.* 358 (2019) 362–368.
- [35] B. Gao, Y. Huang, S. Wang, H. Lu, L. Zheng, X. Fan, X. Yang, W. Zhang, Y. Wang, Y. Zhang, Q. Gao, Y. Tang, MoC nanodots toward efficient electrocatalytic hydrogen evolution: an interlayer-confined strategy with a 2D-zeolite precursor, *J. Mater. Chem. A* 9 (2021) 4724–4733.
- [36] X. Huang, J. Wang, H. Bing Tao, H. Tian, Z. Zhang, H. Xu, Unraveling the oxide layer on Mo₂C as the active center for hydrogen evolution reaction, *J. Catal.* 389 (2020) 461–467.
- [37] J. Jia, T. Xiong, L. Zhao, F. Wang, H. Liu, R. Hu, J. Zhou, W. Zhou, S. Chen, Ultrathin N-doped Mo₂C nanosheets with exposed active sites as efficient electrocatalyst for hydrogen evolution reactions, *ACS Nano* 11 (2017) 12509–12518.
- [38] J. Chen, H. Zhang, J. Yu, D. Guan, S. She, W. Zhou, Z. Shao, Self-catalyzed formation of strongly interconnected multiphase molybdenum-based composites for efficient hydrogen evolution, *Carbon Energy* 4 (2021) 77–87.
- [39] Y. Sun, C. Huang, J. Shen, Y. Zhong, J. Ning, Y. Hu, One-step construction of a transition-metal surface decorated with metal sulfide nanoparticles: a high-efficiency electrocatalyst for hydrogen generation, *J. Colloid Interface Sci.* 558 (2020) 1–8.
- [40] L. Zhao, H. Yuan, D. Sun, J. Jia, J. Yu, X. Zhang, X. Liu, H. Liu, W. Zhou, Active facet regulation of highly aligned molybdenum carbide porous octahedrons via crystal engineering for hydrogen evolution reaction, *Nano Energy* 77 (2020).